with at least an untraveled edge), and choose a different path.

This is the idea behind *Depth First Search* (*DFS*), basically the opposite strategy with respect to BFS; this algorithm, described in listing 14.4, can't be used to find shortest paths, but it has numerous important applications in graph theory, and also in practice.

Listing 14.4 The dfs visit of a vertex (and its neighborhood)

```
function dfs(graph, v, time=0, in_time={}, out_time=
{})                          #1
  time ← time +
1
#2
  in_time[v] ←
time
#3
  for e in graph.adjacencyList[v]
do                                              #4
    u ← e.dest
    if in_time[u] == null
then                                                  #5
      (time, in_time, out_time) ← dfs(graph, u, time, in_time,
out_time)      #6
  time ← time +
1
#7
  out_time[v] ←
time
#8
  return (time, in_time,
out_time)                                               #9
```